

Lab 1: Packet analysis at application layer using Wireshark

SCSR1213 Network Communications

Universiti Teknologi Malaysia

Objective:

1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

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PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

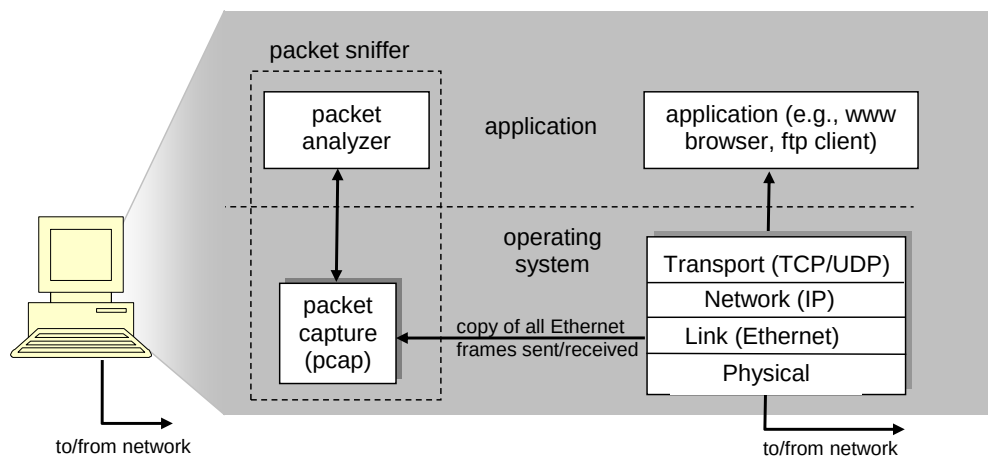


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

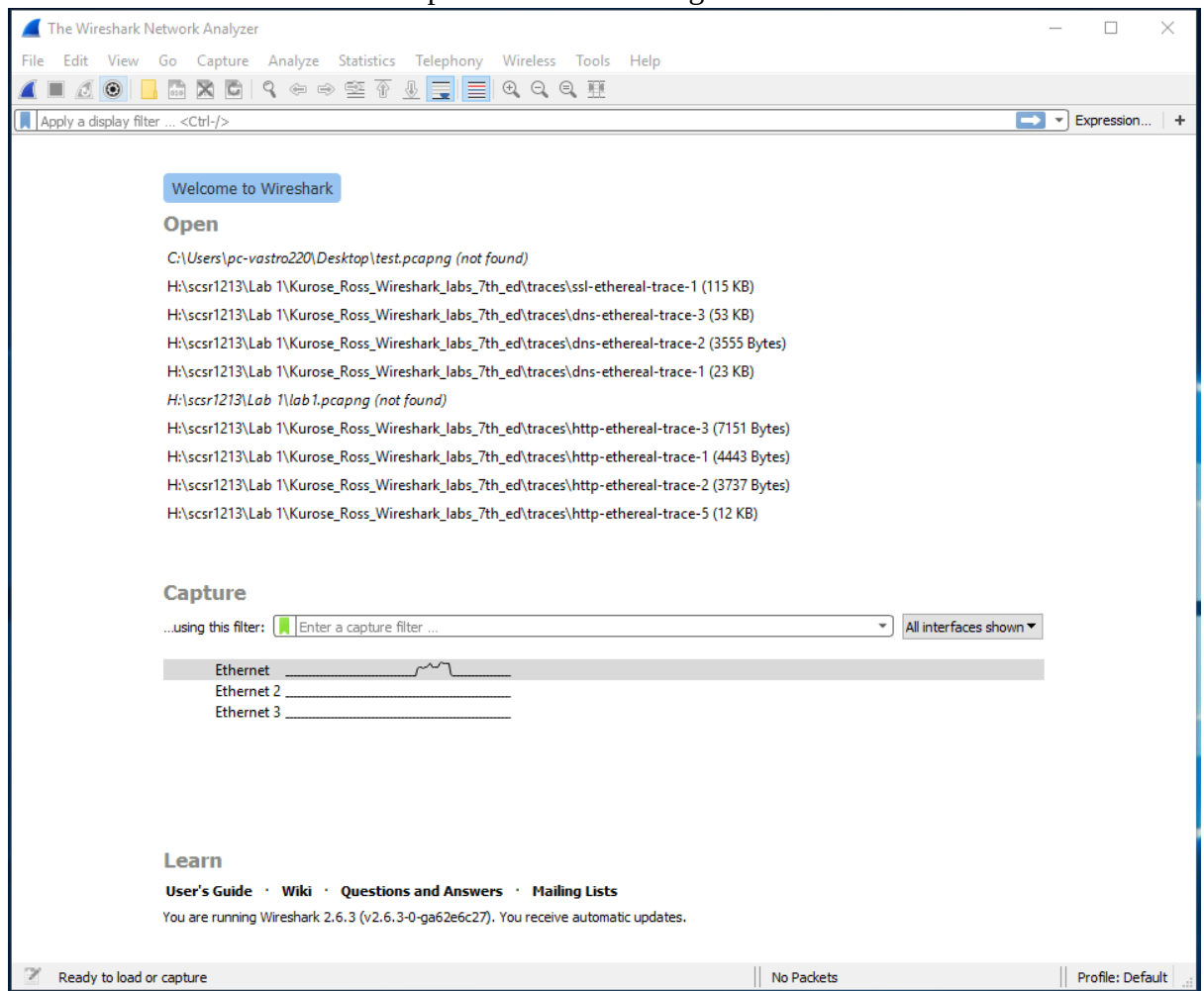


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

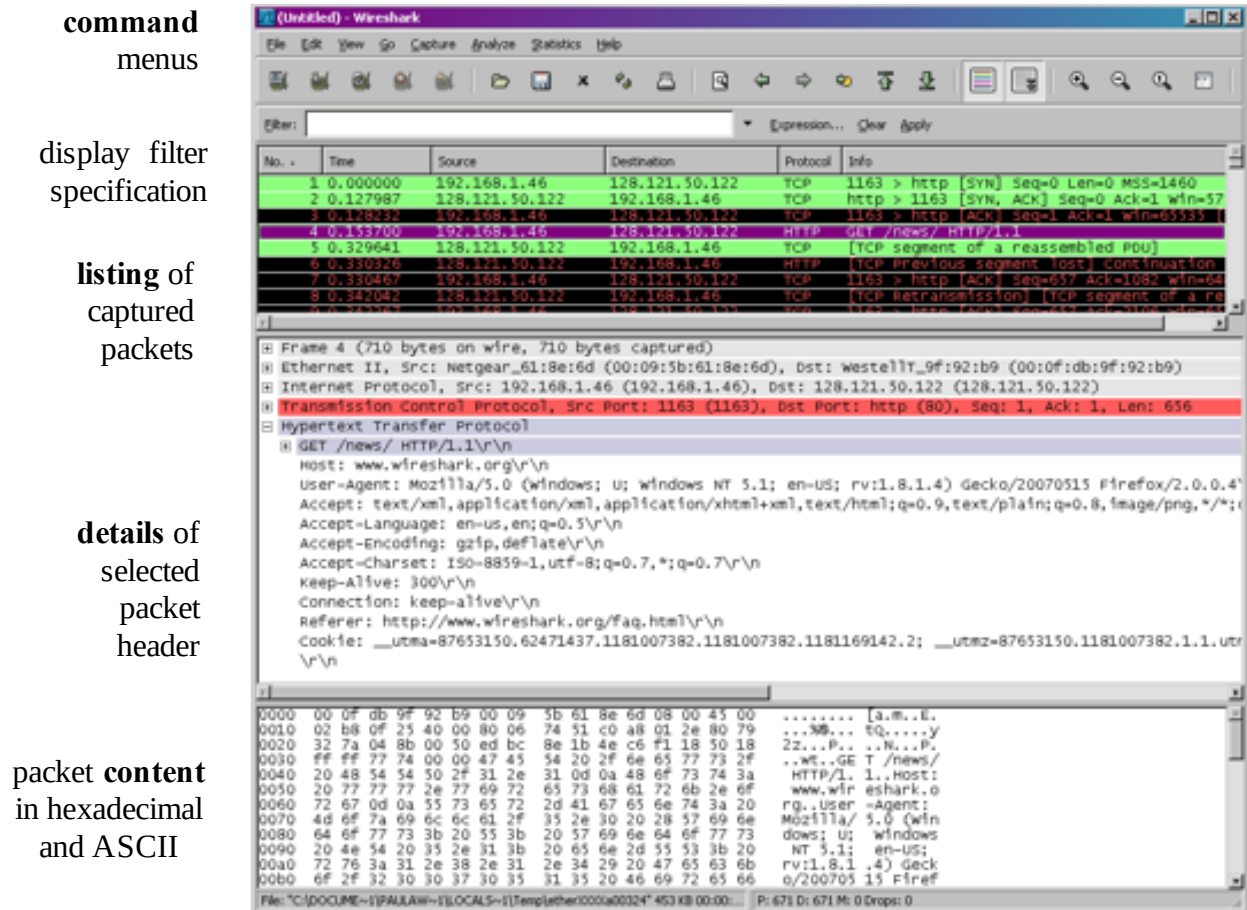


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** displays details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

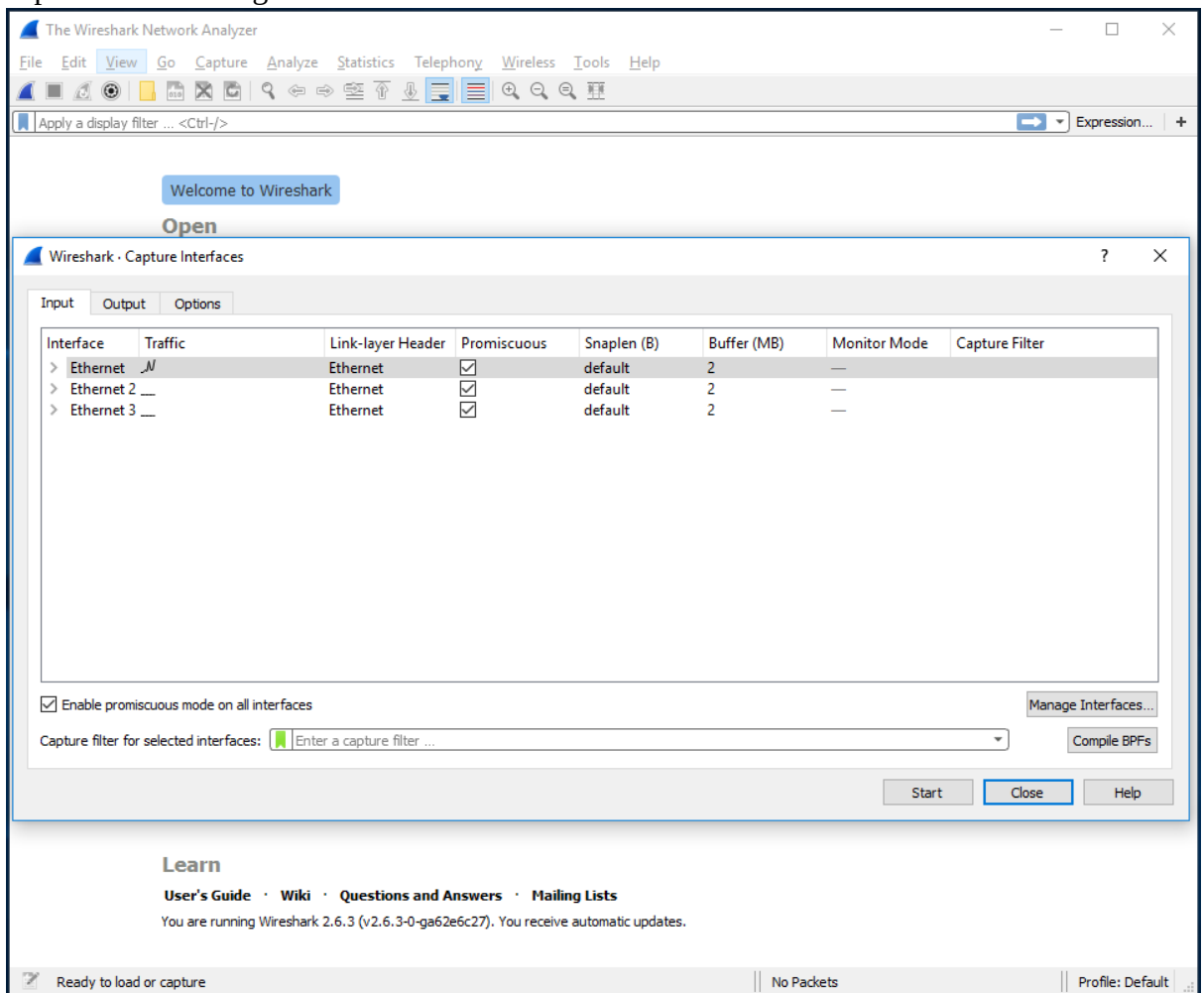


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

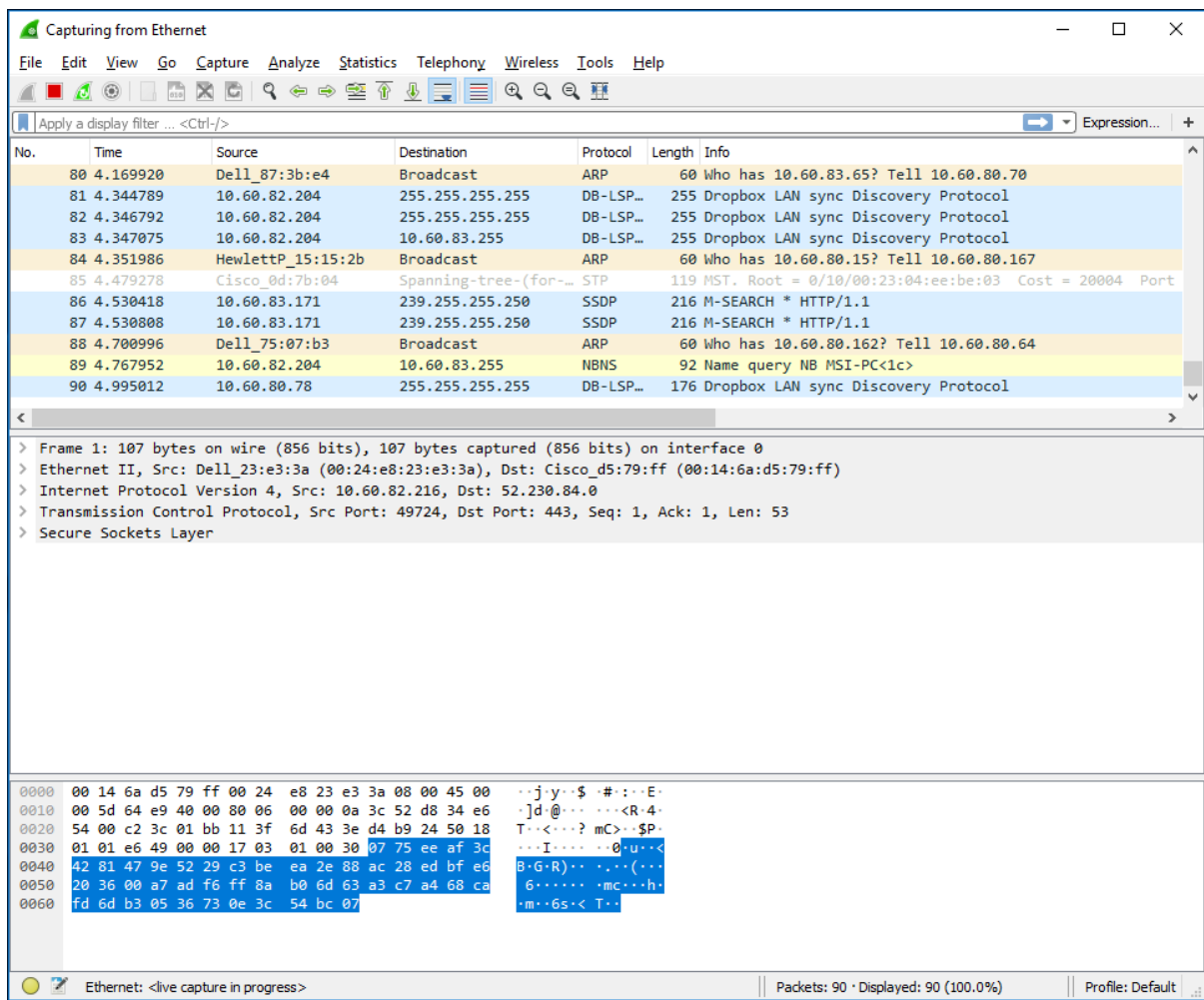


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

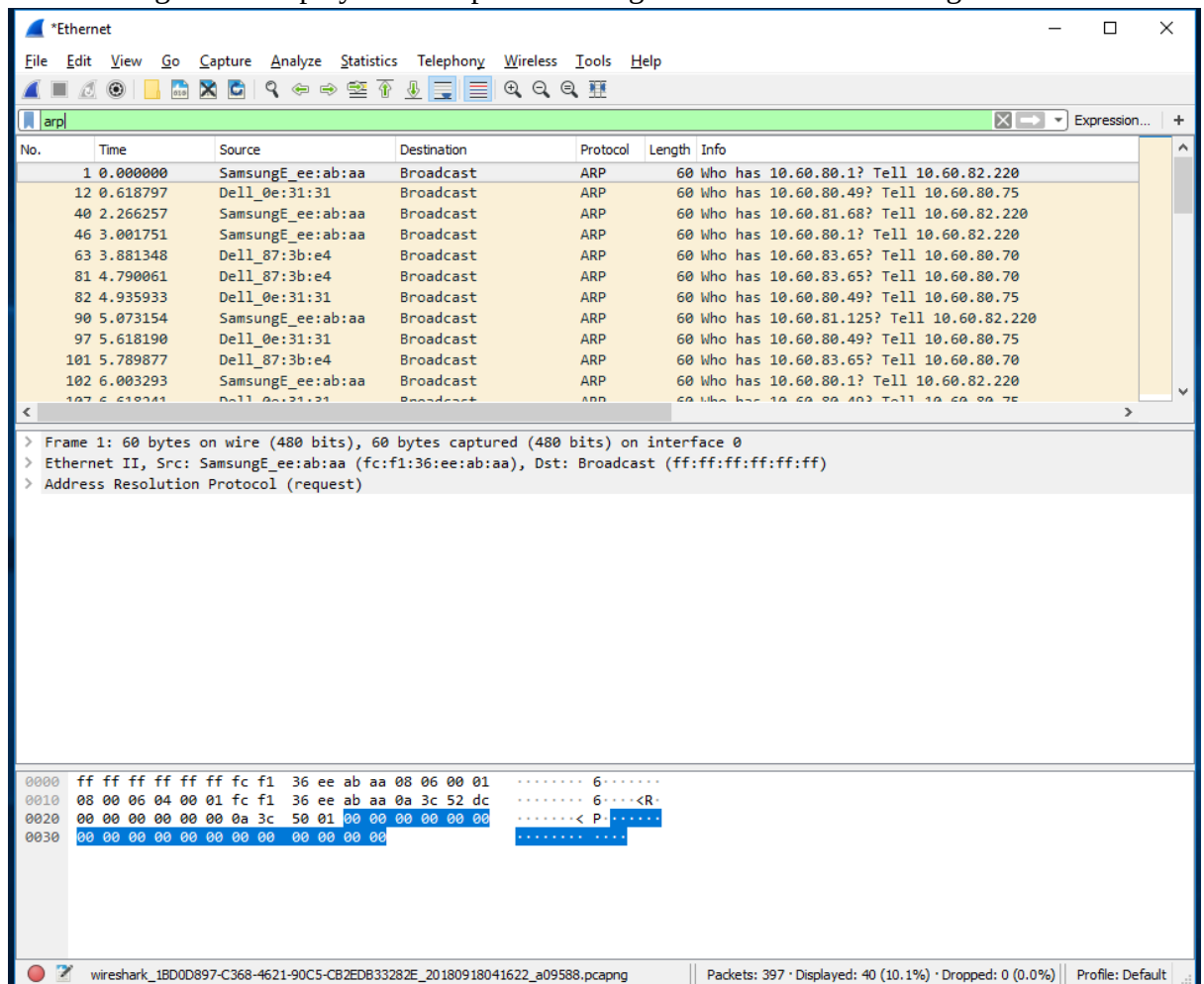


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

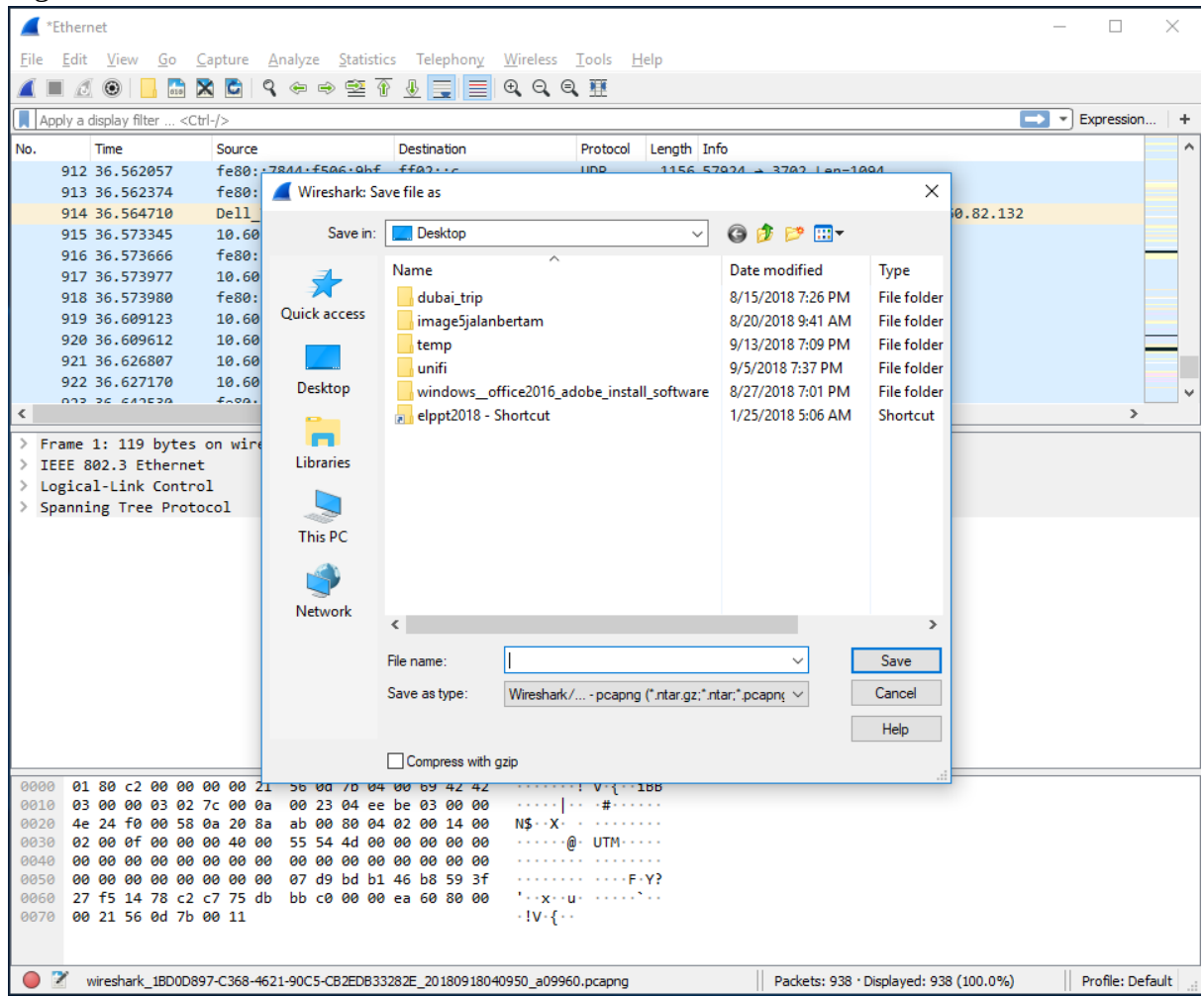


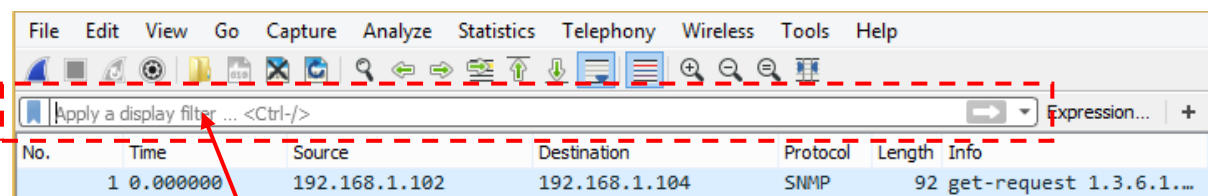
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:
 1. What version of HTTP is the server running?
The server is running HTTP/1.1
 2. What is the IP address of the client computer?
The client's IP address is 128.119.245.12
 3. What is the IP address of the gaia.cs.umass.edu server?
The server's IP address is 192.168.1.102
 4. How many bytes of content are being returned to client browser?
439 bytes and 1395 bytes
 5. What is the status code returned from the server to client browser?
**200 OK for successful response (in line 2),
and 404 NOT FOUND for unsuccessful response (in line 4)**

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

The first HTTP GET request does not include any “IF-MODIFIED-SINCE” line. It just requests for the file /ethereal-labs/lab2-2.html

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

Yes, the server does explicitly return the contents of the file. This can be told from the response, 200 OK (text/html).

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes, I did see an “IF-MODIFIED-SINCE” line in the HTTP GET. The followed information is (Tue, 23 Sep 2003 05:35:00 GMT\r\n) .

```
Connection: keep-alive\r\n
If-Modified-Since: Tue, 23 Sep 2003 05:35:00 GMT\r\n
If-None-Match: "1bfef-173-8f4ae900"\r\n
Cache-Control: max-age=0\r\n
\r\n
```

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file?

Explain.

13	3.033773	192.168.1.104	192.168.1.102	SNMP	93	get-response 1.3.6.1.4.1.11.2.3.9.4.2.1.2.2.1.0
14	5.517390	192.168.1.102	128.119.245.12	HTTP	668	GET /ethereal-labs/lab2-2.html HTTP/1.1
15	5.540216	128.119.245.12	192.168.1.102	HTTP	243	HTTP/1.1 304 Not Modified
16	5.658309	192.168.1.102	128.119.245.12	TCP	54	4247 → 80 [ACK] Seq=1116 Ack=875 Win=63366 Len=0
17	6.033489	192.168.1.102	192.168.1.104	SNMP	92	get-request 1.3.6.1.4.1.11.2.3.9.4.2.1.2.2.1.0

The response is 304 NOT MODIFIED, means that the server does not explicitly return the contents of the file. This response indicates the requested resource has not changed since the last request, and the server tells the client to use the cached version instead.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?
3 request messages.

2. To which Internet addresses were these GET requests sent?

10	7.236929	192.168.1.102	128.119.245.12	HTTP	555 GET /ethereal-labs/lab2-4.html HTTP/1.1
12	7.260813	128.119.245.12	192.168.1.102	HTTP	1057 HTTP/1.1 200 OK (text/html)
17	7.305485	192.168.1.102	165.193.123.218	HTTP	625 GET /catalog/images/pearson-logo-footer.gif HTTP/1.1
20	7.308803	192.168.1.102	134.241.6.82	HTTP	609 GET /~kurose/cover.jpg HTTP/1.1
25	7.333054	165.193.123.218	192.168.1.102	HTTP	912 HTTP/1.1 200 OK (GIF89a)
54	7.589877	134.241.6.82	192.168.1.102	HTTP	1096 HTTP/1.0 200 Document follows (JPEG JFIF image)

The address is 128.119.245.12, 165.193.123.218, and 134.241.6.82

3. How many bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?
912 bytes.

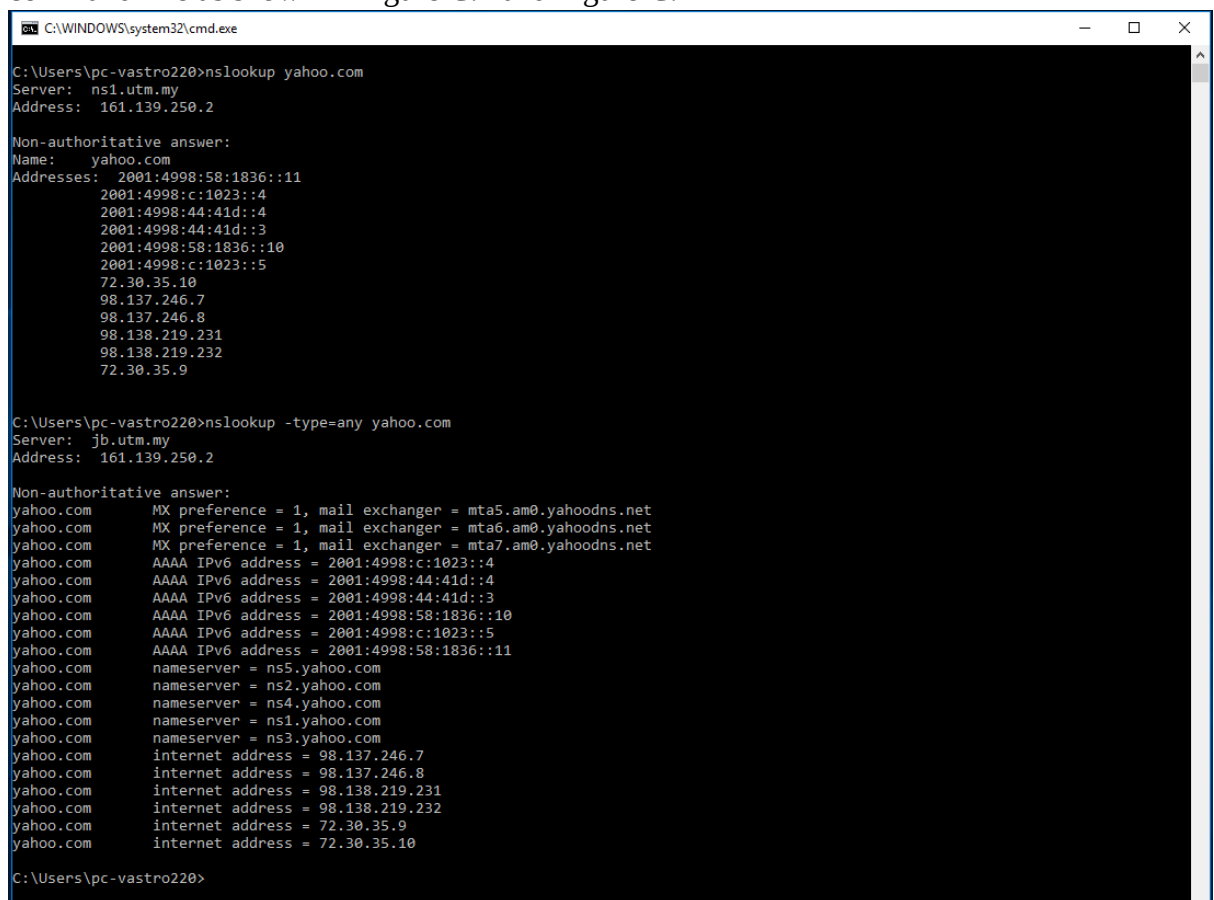
4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?
1096 bytes.

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

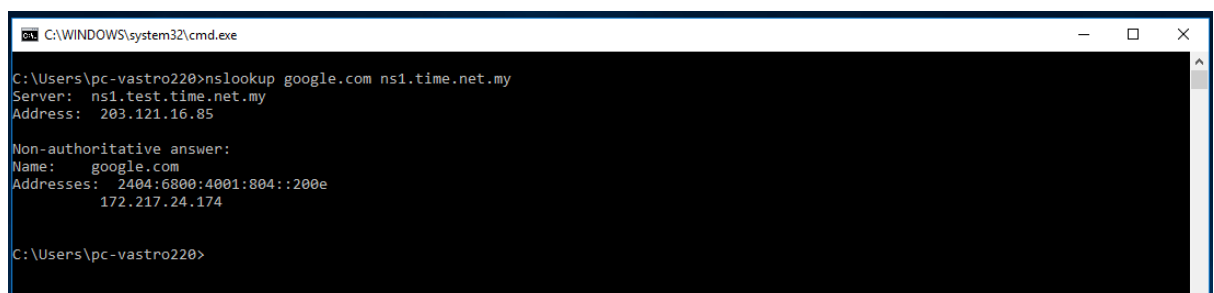
Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Users\Asus>nslookup www.microsoft.com
Server:    ns1.utm.my
Address:   161.139.250.2

Non-authoritative answer:
Name:      e13678.dscb.akamaiedge.net
Addresses: 2001:f40:a:c82::356e
           2001:f40:a:c84::356e
           2001:f40:a:c87::356e
           2001:f40:a:c8e::356e
           2001:f40:a:c81::356e
           211.25.121.218
Aliases:   www.microsoft.com
           www.microsoft.com-c-3.edgekey.net
           www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

The IP address of the server is 211.25.121.218

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

```
C:\Users\Asus>nslookup www.microsoft.com
Server:    ns1.utm.my
Address:   161.139.250.2

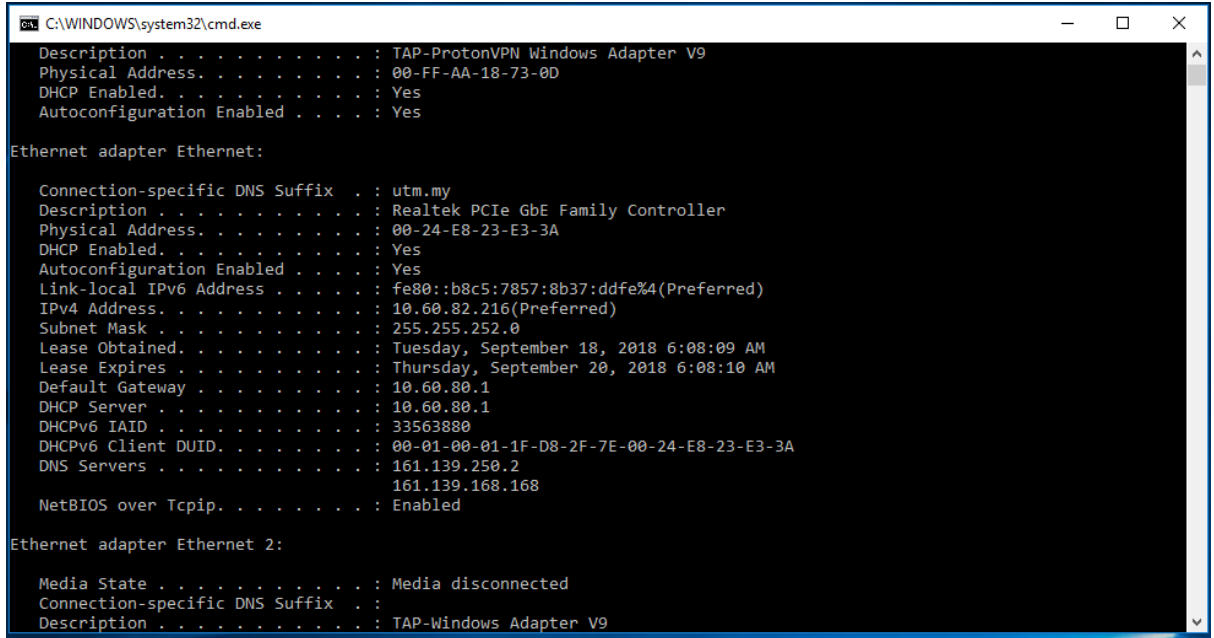
Non-authoritative answer:
Name:      e13678.dscb.akamaiedge.net
Addresses: 2001:f40:a:c82::356e
           2001:f40:a:c84::356e
           2001:f40:a:c87::356e
           2001:f40:a:c8e::356e
           2001:f40:a:c81::356e
           211.25.121.218
Aliases:   www.microsoft.com
           www.microsoft.com-c-3.edgekey.net
           www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

The IP address of the non-authoritative DNS servers for domain miscrosoft.com are 2001:f40:a:c82::356e, 2001:f40:c84::356e, 2001:f40:a:c87::356e, 2001:f40:a:c8e::356e, 2001:f40:a:c81::356e

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

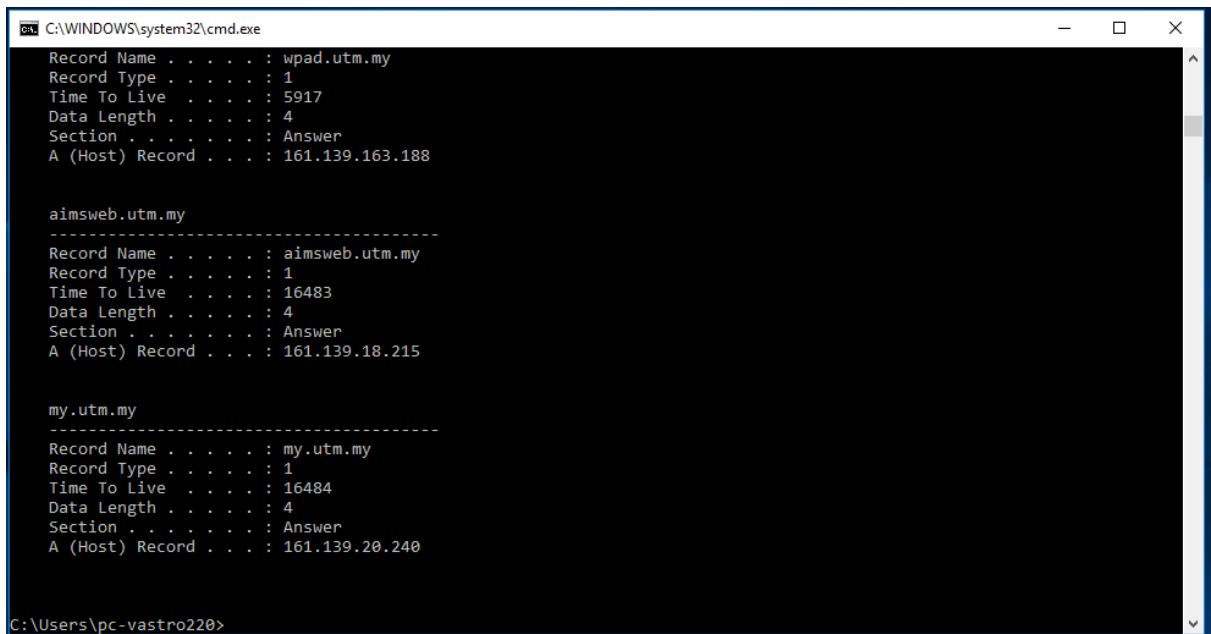
    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds. Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
aimsweb.utm.my
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

-----
my.utm.my
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns



```
C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

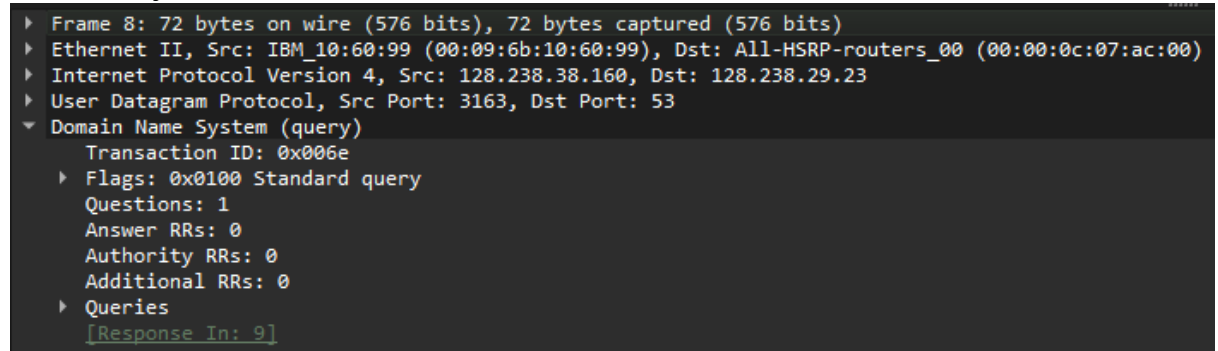
C:\Users\pc-vastro220>
```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

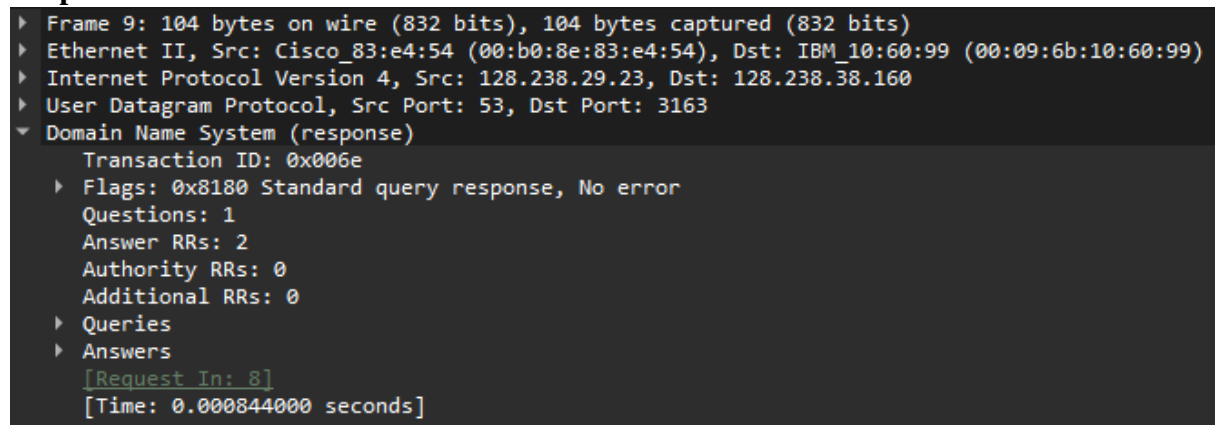
- Open packet trace file dns-trace-1. Answer the following questions.
- 1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

DNS Query:



```
▶ Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
▶ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
▶ User Datagram Protocol, Src Port: 3163, Dst Port: 53
▼ Domain Name System (query)
  Transaction ID: 0x006e
  ▶ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▶ Queries
    [Response In: 9]
```

Response:



```
▶ Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
▶ Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
▶ Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
▶ User Datagram Protocol, Src Port: 53, Dst Port: 3163
▼ Domain Name System (response)
  Transaction ID: 0x006e
  ▶ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  ▶ Queries
  ▶ Answers
    [Request In: 8]
    [Time: 0.000844000 seconds]
```

They are sent over UDP

2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

DNS Query:

```
▶ Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-MSRP-routers_00 (00:00:0c:07:ac:00)
▶ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
▼ User Datagram Protocol, Src Port: 3163, Dst Port: 53
    Source Port: 3163
    Destination Port: 53
    Length: 38
    Checksum: 0x8acb [unverified]
    [Checksum Status: Unverified]
    [Stream index: 1]
    [Stream Packet Number: 1]
    ▶ [Timestamps]
    UDP payload (30 bytes)
▶ Domain Name System (query)
```

Response:

```
▶ Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
▶ Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
▶ Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
▼ User Datagram Protocol, Src Port: 53, Dst Port: 3163
    Source Port: 53
    Destination Port: 3163
    Length: 70
    Checksum: 0xb0ba [unverified]
    [Checksum Status: Unverified]
    [Stream index: 1]
    [Stream Packet Number: 2]
    ▶ [Timestamps]
    UDP payload (62 bytes)
▶ Domain Name System (response)
```

Both destination port of DNS query and source port of DNS response message are 53.

3. To what IP address is the DNS query message sent? Add screenshots in your answer.

```
▶ Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
▼ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    ▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
        Total Length: 58
        Identification: 0x229e (8862)
    ▶ 000. .... = Flags: 0x0
        ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0xd281 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 128.238.38.160
    Destination Address: 128.238.29.23
    [Stream index: 1]
▶ User Datagram Protocol, Src Port: 3163, Dst Port: 53
▶ Domain Name System (query)
```

The destination address is 128.238.29.23

4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

```
▶ Frame 8: 72 bytes on wire (576 bits), 72 bytes captured (576 bits)
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
▶ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
▶ User Datagram Protocol, Src Port: 3163, Dst Port: 53
▼ Domain Name System (query)
    Transaction ID: 0x006e
    ▶ Flags: 0x0100 Standard query
        Questions: 1
        Answer RRs: 0
        Authority RRs: 0
        Additional RRs: 0
    ▼ Queries
        ▶ www.ietf.org: type A, class IN
        [Response In: 9]
```

The type of DNS Query is type A, there are no “answers”.

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

```

▶ Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
▶ Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
▶ Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
▶ User Datagram Protocol, Src Port: 53, Dst Port: 3163
▼ Domain Name System (response)
  Transaction ID: 0x006e
  ▶ Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 2
  Authority RRs: 0
  Additional RRs: 0
  ▼ Queries
    ▶ www.ietf.org: type A, class IN
      Name: www.ietf.org
      [Name Length: 12]
      [Label Count: 3]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
  ▼ Answers
    ▶ www.ietf.org: type A, class IN, addr 132.151.6.75
    ▶ www.ietf.org: type A, class IN, addr 65.246.255.51
    [Request In: 8]
    [Time: 0.000844000 seconds]

```

There are 2 “answers” (Answers RRs: 2).

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

TCP SYN packet:

```

▶ Frame 10: 62 bytes on wire (496 bits), 62 bytes captured (496 bits)
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
▶ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 132.151.6.75
▶ Transmission Control Protocol, Src Port: 3369, Dst Port: 80, Seq: 0, Len: 0

```

DNS response:

Time	Source	Destination	Protocol	Length	Info
8.3075845	128.238.38.160	128.238.29.23	DNS	72	Standard query 0x006e A www.ietf.org
9.3076689	128.238.29.23	128.238.38.160	DNS	104	Standard query response 0x006e A www.ietf.org A 132.151.6.75 A 65.246.255.51


```

▶ Frame 9: 104 bytes on wire (832 bits), 104 bytes captured (832 bits)
▶ Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
▶ Internet Protocol Version 4, Src: 128.238.29.23, Dst: 128.238.38.160
▶ User Datagram Protocol, Src Port: 53, Dst Port: 3163
▼ Domain Name System (response)
  Transaction ID: 0x006e
  ▶ Flags: 0x8180 Standard query response, No error
  Questions: 1
  Answer RRs: 2
  Authority RRs: 0
  Additional RRs: 0
  ▼ Queries
    ▶ www.ietf.org: type A, class IN
      Name: www.ietf.org
      [Name Length: 12]
      [Label Count: 3]
      Type: A (1) (Host Address)
      Class: IN (0x0001)
  ▼ Answers
    ▶ www.ietf.org: type A, class IN, addr 132.151.6.75
    ▶ www.ietf.org: type A, class IN, addr 65.246.255.51
    [Request In: 8]
    [Time: 0.000844000 seconds]

```

Yes, the IP address is 132.151.6.75

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

13	3.096708	128.238.38.160	132.151.6.75	HTTP	429 GET / HTTP/1.1
20	3.153211	132.151.6.75	128.238.38.160	HTTP	1055 HTTP/1.1 200 OK (text/html)
28	3.191998	128.238.38.160	132.151.6.75	HTTP	320 GET /images/ietflogo2e.gif HTTP/1.1
31	3.192869	128.238.38.160	132.151.6.75	HTTP	314 GET /images/blue.gif HTTP/1.1
36	3.228451	132.151.6.75	128.238.38.160	HTTP	1212 HTTP/1.1 200 OK (GIF89a)
39	3.230578	132.151.6.75	128.238.38.160	HTTP	407 HTTP/1.1 200 OK (GIF89a)
52	3.286844	128.238.38.160	132.151.6.75	HTTP	317 GET /images/redstar.gif HTTP/1.1
53	3.287024	128.238.38.160	132.151.6.75	HTTP	319 GET /images/blue-line.jpg HTTP/1.1
55	3.309302	132.151.6.75	128.238.38.160	HTTP	1215 HTTP/1.1 200 OK (GIF89a)
61	3.321019	132.151.6.75	128.238.38.160	HTTP	741 HTTP/1.1 200 OK (JPEG JFIF image)
71	3.353822	128.238.38.160	132.151.6.75	HTTP	320 GET /images/isoc-small.gif HTTP/1.1
75	3.384003	132.151.6.75	128.238.38.160	HTTP	1269 HTTP/1.1 200 OK (GIF87a)

No.

- Open packet trace file dns-trace-2 for nslookup.
- We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
- Answer the following questions.

8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

DNS Query:

```

▶ Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits)
▶ Ethernet II, Src: IBM_10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)
▶ Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22
▶ User Datagram Protocol, Src Port: 3742, Dst Port: 53
▶ Domain Name System (query)

```

DNS Response:

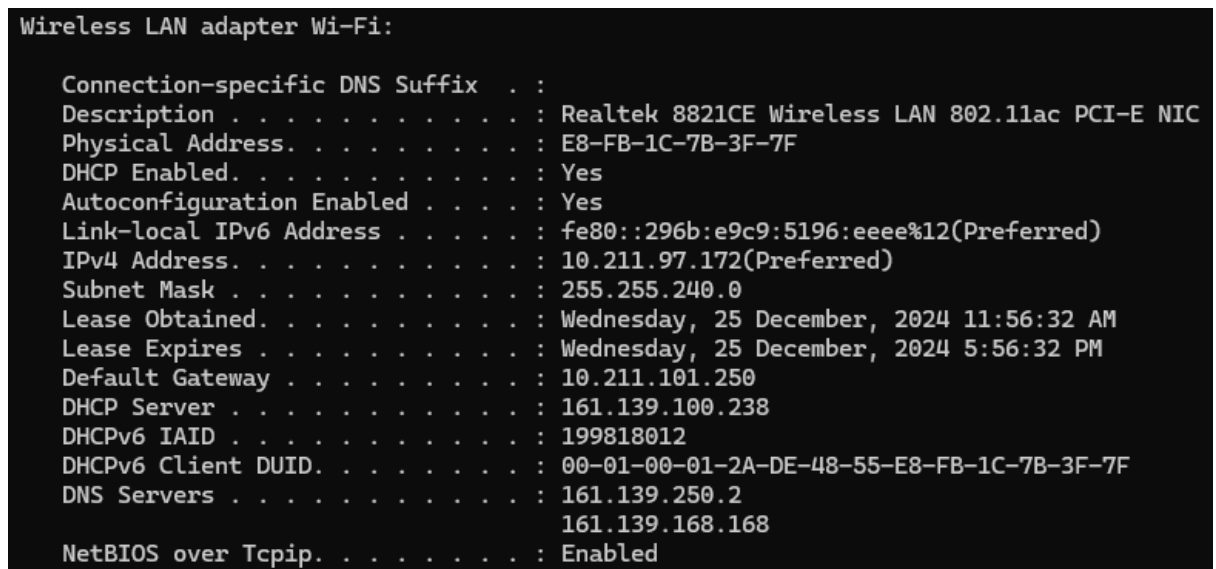
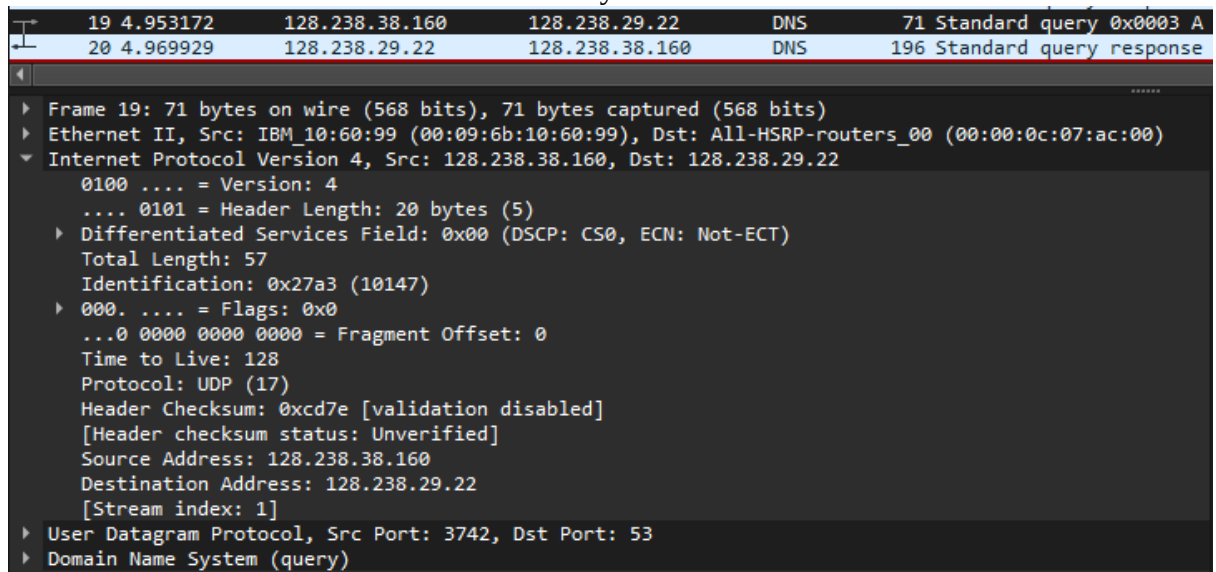
```

▶ Frame 20: 196 bytes on wire (1568 bits), 196 bytes captured (1568 bits)
▶ Ethernet II, Src: Cisco_83:e4:54 (00:b0:8e:83:e4:54), Dst: IBM_10:60:99 (00:09:6b:10:60:99)
▶ Internet Protocol Version 4, Src: 128.238.29.22, Dst: 128.238.38.160
▶ User Datagram Protocol, Src Port: 53, Dst Port: 3742
▶ Domain Name System (response)

```

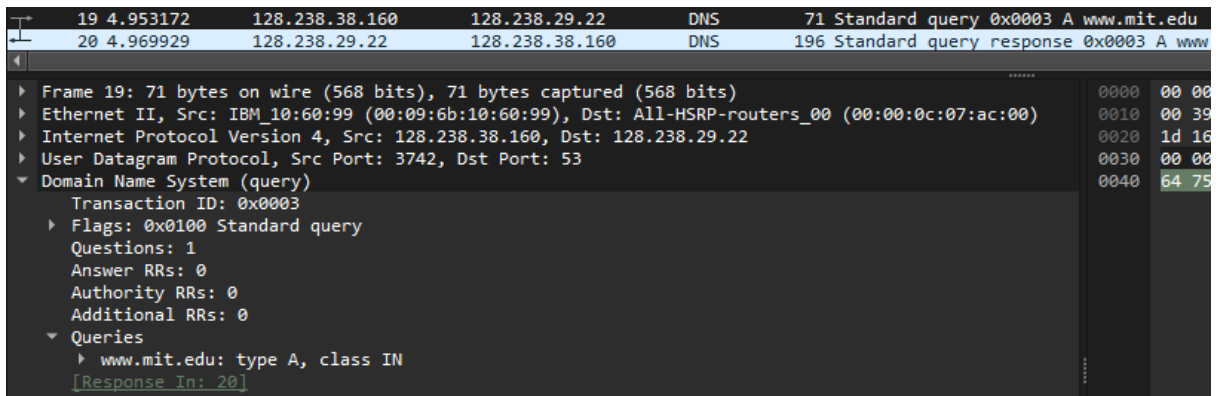
Both destination port for the DNS query message and source port of DNS response message are 53.

9. To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.



The DNS query message is sent to the IP address of 128.238.29.22. This is not the IP address of my default local DNS server as above.

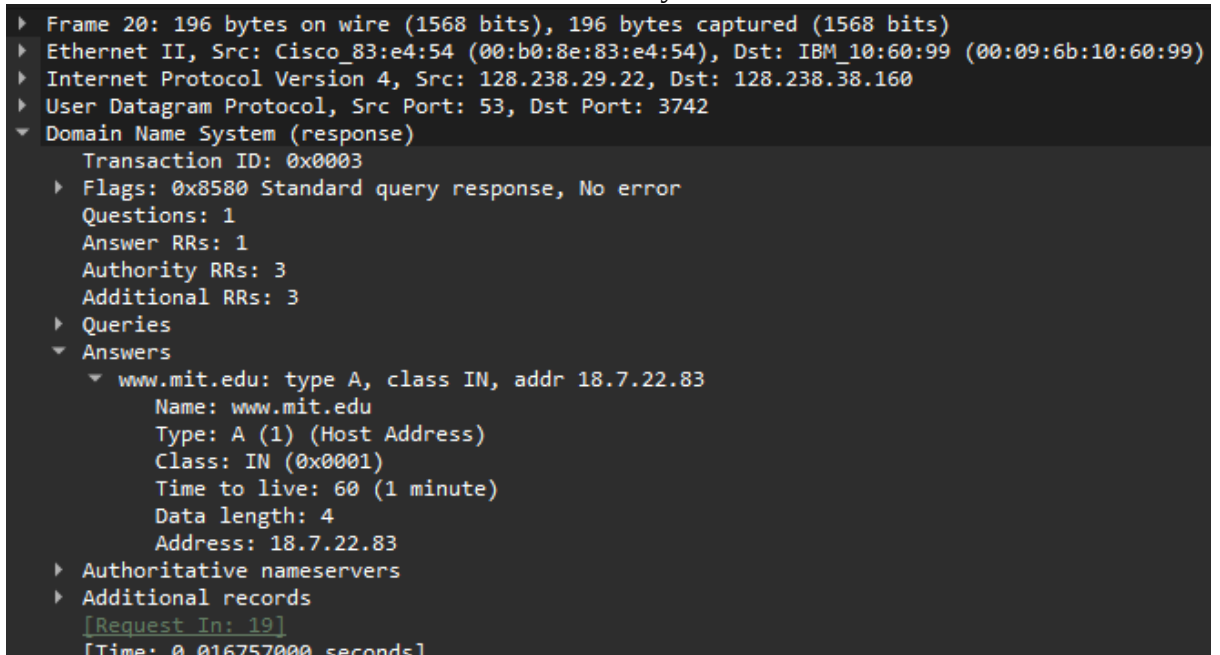
10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.



The screenshot shows a Wireshark packet capture of a DNS query. The packet list at the top shows two packets: packet 19 (71 bytes) and packet 20 (196 bytes). Packet 19 is a 'Standard query' from 128.238.38.160 to 128.238.29.22. Packet 20 is a 'Standard query response' from 128.238.29.22 to 128.238.38.160. The packet details for packet 19 are expanded, showing the 'Domain Name System (query)' section. It includes the transaction ID 0x0003, flags 0x0100 (Standard query), one question, and no answers, authority records, or additional records. The question is for 'www.mit.edu: type A, class IN'. The packet bytes pane on the right shows the raw data in hexadecimal and ASCII.

Type A, does not contain any “answers”.

11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.



The screenshot shows a Wireshark packet capture of a DNS response. The packet details for packet 20 are expanded, showing the 'Domain Name System (response)' section. It includes the transaction ID 0x0003, flags 0x8580 (Standard query response, No error), one question, one answer, three authority records, and three additional records. The answer is for 'www.mit.edu: type A, class IN, addr 18.7.22.83'. The details of the answer are expanded, showing the name 'www.mit.edu', type 'A (1) (Host Address)', class 'IN (0x0001)', time to live '60 (1 minute)', data length '4', and address '18.7.22.83'. The packet bytes pane on the right shows the raw data in hexadecimal and ASCII.

One “answer” is provided:

- **Domain name being queried**
- **Type of address**
- **The “Internet” class (0x0001 is the hexadecimal value for the Internet class)**
- **Time to live: 60 (1 minute) - indicates how long the record can be cached. The DNS resolver should request a fresh copy after 60 seconds.**
- **Data length: 4 - indicates the length of the address data in bytes.**
- **Address: 18.7.22.83 - the actual IPv4 address that www.mit.edu resolves to**